Introduction to Drone Technology (IDT)
Assignment 1: Design the Drone for a Mission

Deadline: 1st September 2026, 2 PM

Task: You are a drone technology startup. A client approaches you with a real-world problem. Your team must design a drone-based solution that can realistically solve it.

Team Formation: Groups of 4 students. Each group will receive a different mission.

Team	Challenge
1	Identify crop stress and monitor agricultural fields.
2	Detect and monitor forest fires.
3	Deliver medicines to remote villages.
4	Locate missing people after a disaster.
5	Monitor construction progress.
6	Monitor traffic and detect accidents.
7	Map flooded areas and identify stranded people.
8	Inspect electrical transmission lines.
9	Deliver small packages.
10	Solve one urban problem using UAVs.

1. Mission Definition

- What problem are you solving and who is potential user/customer?
- Choose a baseline existing solution and justify why the proposed solution/drone is better.

2. Design Your Drone

- Select the configuration: fixed-wing, multirotor, VTOL or hybrid.
- Determine number of motors/propulsion units and approximate size.
- Specify payload, battery/power source and expected flight time.
- Define operating altitude and communication range.
- Select navigation and control approach.

3. Select Technology

- Flight controller, IMU, GPS/GNSS, Barometer, Magnetometer, Camera and/or mission-specific sensors, ESCs, Motors and propellers, Battery, Communication/data link

4. Engineering Calculations

- Estimated take-off weight (MTOW)
- Required thrust and thrust-to-weight ratio
- Battery capacity and estimated flight time
- Propeller/motor selection
- Payload capacity
- Wing loading for fixed-wing UAVs
- Estimated range

- Power consumption

5. Drone Identity Card

Prepare a one-page identity card for your proposed drone containing:

- Drone name
- Mission
- Configuration
- MTOW
- Payload
- Endurance
- Sensors
- Communication range
- Estimated cost
- Unique feature

6. Pitch Your Drone

Each team gets 10 minutes to present the idea. The presentation will be followed by 2 minutes of questions from the class.

8. Evaluation Scheme – 20 Marks

Evaluation Component	**Marks**
Problem definition & mission	2
Drone configuration selection	3
Component selection	3
Engineering calculations	4
Feasibility & cost	3
Creativity/innovation	2
Poster/design quality	1
Presentation & Q&A	2
Total	20

Required Submission

- Drone Identity Card
- Technical report including engineering calculation sheet
- 10-minute presentation